

0 1 . 1

Describe how ultrafiltration occurs in a glomerulus.

[3 marks]

0 1 . 2

Glucose and water are reabsorbed by the proximal convoluted tubule of a nephron.

Put a tick (✓) in the box next to the correct ways in which glucose and water are reabsorbed.

[1 mark]

Glucose by active transport and water against a water potential gradient

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Glucose by diffusion and water down a water potential gradient

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Glucose by facilitated diffusion and active transport and water against a water potential gradient

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Glucose by facilitated diffusion and active transport and water down a water potential gradient

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The equation shows the relationship between urine concentration in arbitrary units (y) and mean length of the loop of Henle in mm (x).

$$y = 0.72x + 4$$

Calculate the mean length of the loop of Henle in an organism that produces urine with a concentration of 16.56 arbitrary units.

[1 mark]

Answer = _____ mm

Question 1 continues on the next page
